What is EUV?

Extreme Ultraviolet

Lithography that enables smaller semiconductor features

EUV is extreme ultraviolet light with a wavelength of 13.5 nm. It is used in lithography to transfer tiny circuit patterns onto semiconductor wafers.

DUV and EUV compared

Technology	Wavelength and uses
DUV (deep ultraviolet)	ArF sources: 193 nm. Used in the manufacture of a broad range of semiconductors.
EUV (extreme ultraviolet)	13.5 nm. Used for advanced logic chips and some advanced DRAM.

Why EUV matters

Miniaturization: The shorter wavelength makes it easier to form finer patterns.
Fewer process steps: On some layers, EUV reduces the need for multiple exposures.
Technology barriers: Light sources, mirrors, masks and inspection require highly advanced technologies.

The EUV industry

Companies behind lithography, supporting equipment and chip production

ASML / Netherlands

Holds a monopoly position in the commercial supply of EUV lithography systems.

TSMC / Taiwan

Mass-produces leading-edge logic semiconductors.

Samsung Electronics / South Korea

Manufactures advanced logic semiconductors and DRAM.

SK hynix / South Korea

Manufactures advanced DRAM.

Tokyo Electron / Japan

Supplies supporting semiconductor production equipment, including coating and developing systems.

Lasertec / Japan

Supplies inspection systems for EUV masks.